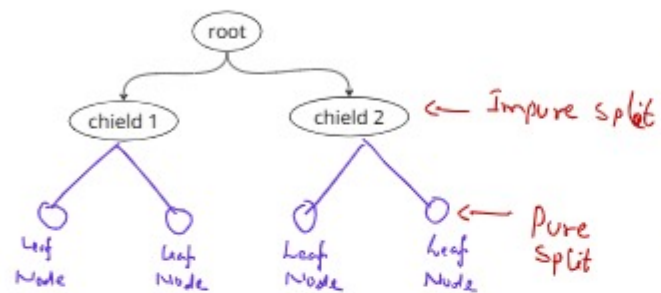
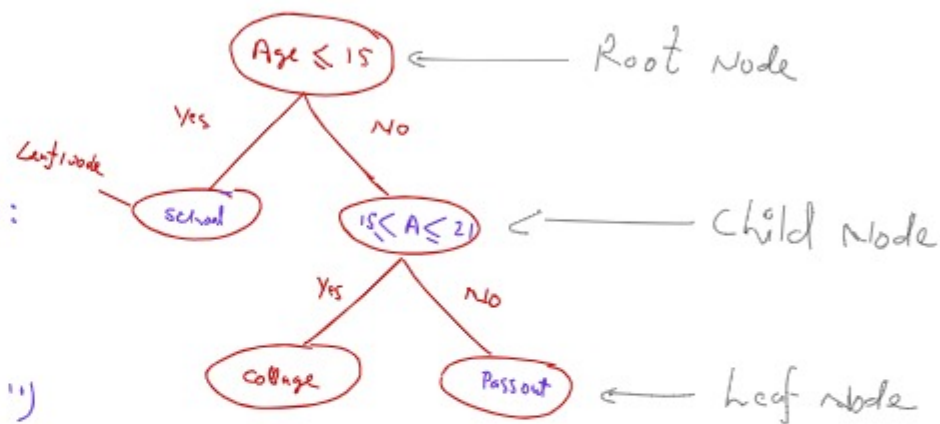


Decision Tree Algorithms

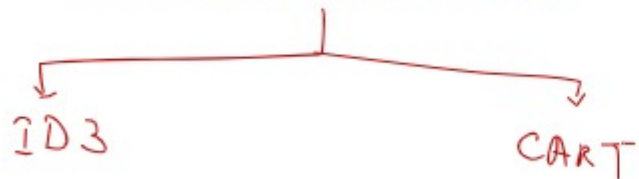


Age = 14

if Age ≤ 15 :
Print ("School")
elif Age ≥ 15 and Age ≤ 21 :
Print ("College")
else:
Print ("College Passout")



Decision Tree Classifier

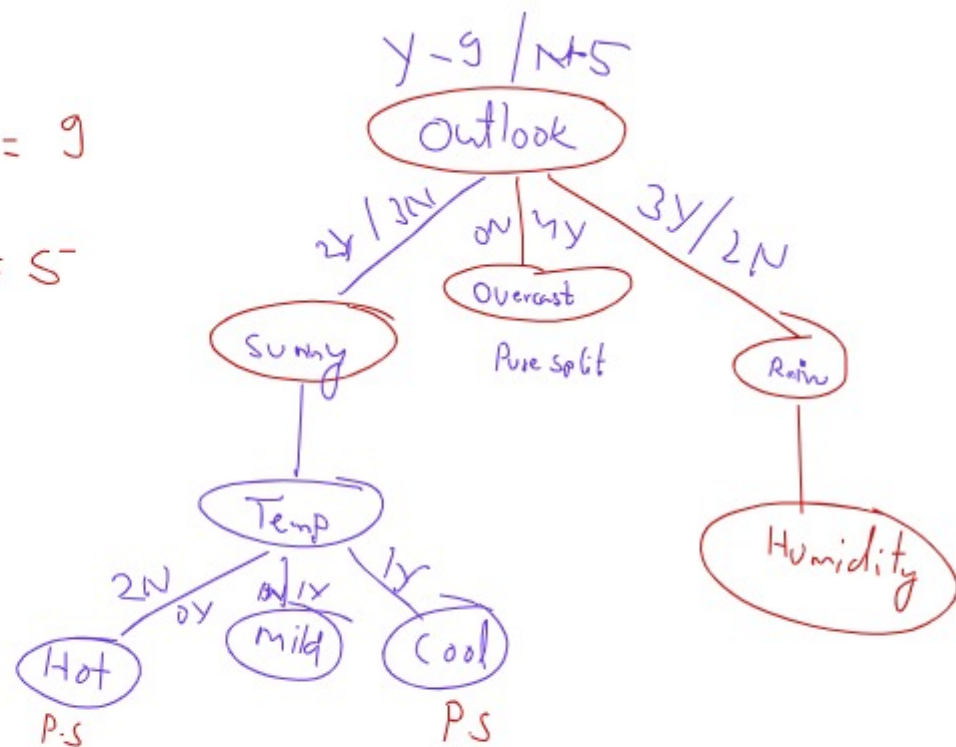


- ① Entropy & Gini Index $\rightarrow$ Purity Split
- ② Information Gain $\rightarrow$ Feature decision tree split

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	False	Yes
6	Rain	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rain	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rain	Mild	High	True	No

Yes = 9

No = 5

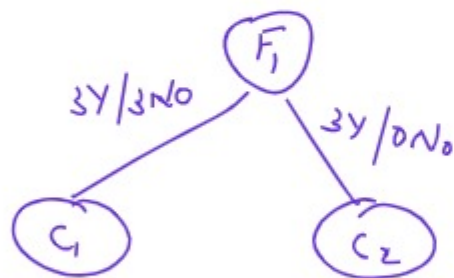


PURITY ---> PURE SPLIT ??

→ Entropy
→ Gini Index / Gini Impurity.

Entropy

$$H(S) = -P_+ \log_2 P_+ - P_- \log_2 P_-$$



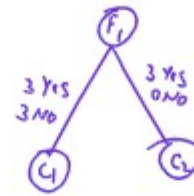
$$C_2 \Rightarrow H(S) = -\frac{3}{3} \log_2 \frac{3}{3} - \frac{0}{3} \log_2 \frac{0}{3} = -1 \log_2 1 = 0 = \text{Pure split}$$

$$\begin{aligned} C_1 \Rightarrow H(S) &= -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = -\frac{1}{2} \left(\log_2 \frac{1}{2} + \log_2 \frac{1}{2} \right) \\ &= -\frac{1}{2} = (-1 + (-1)) = +1 \quad \text{Impure split} \end{aligned}$$

Gini Index / Gini Impurity

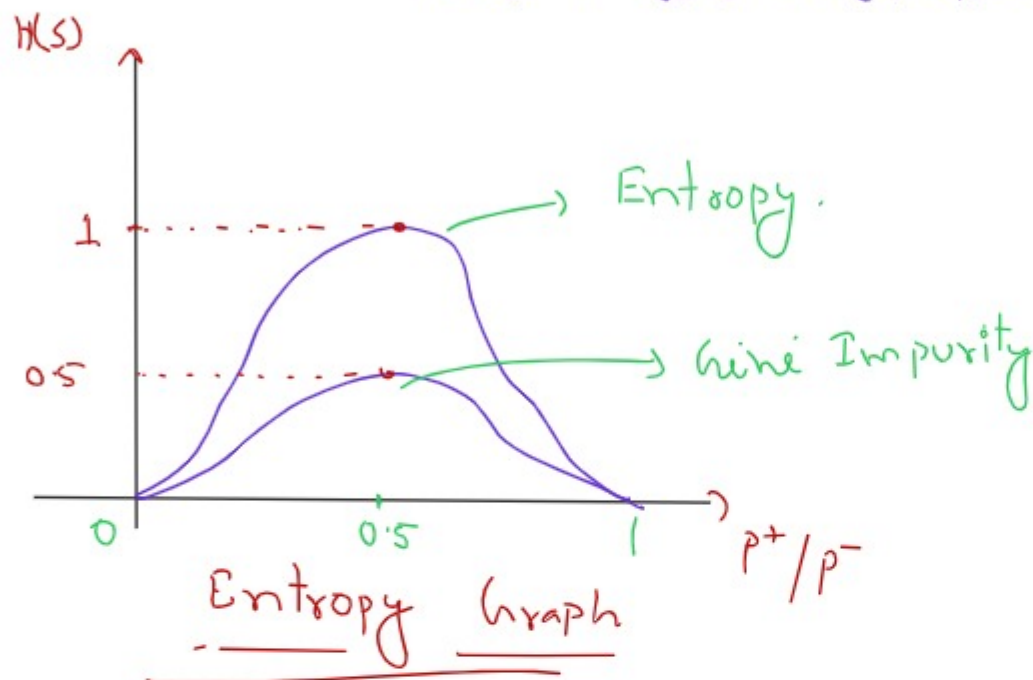
$$G_i = 1 - \sum_{i=1}^n (p_i)^2$$

$$= 1 - \sum_{i=1}^n [(p_+)^2 + (p_-)^2]$$



$$C_2 \rightarrow G_i = 1 - \left[\left(\frac{3}{3}\right)^2 + \left(\frac{0}{3}\right)^2 \right] = 1 - 1 = 0 \text{ Pure Split}$$

$$C_1 \rightarrow G_i = 1 - \left[\left(\frac{3}{6}\right)^2 + \left(\frac{3}{6}\right)^2 \right] = 1 - \left[\frac{1}{4} + \frac{1}{4} \right] = 0.5$$



Information Gain :

$$f_1 \quad f_2 \quad f_3 \quad f_4$$

$$Gini(S|f_i) = H(S) - \sum_{i=1}^n \frac{|Sv|}{|S|} \times H(Sv)$$

$$H(S) = -P_+ \log_2 P_+ - P_- \log_2 P_- \text{ (Entropy of Root Node)}$$

$$H(Sv) = \text{Entropy of Category } C_1, C_2, \dots, C_n.$$

$$Sv = \text{No of Sample of each Node.}$$

$$S = \text{total No of Sample.}$$